Geometric analysis of primate molars - Classifying the diet of Teilhardina

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Purpose

In this project we worked with 3D scan data from 116 primate molars. The focus is on the shape and surface of the teeth, and how this relates to what kind of food the monkey usually ate. Some of the monkeys in the data are extinct, so we don't know their diet. One of them is *Teilhardina*, and in this report we try to guess what it probably ate, based on our classification model.

Each molar is summarized using four numerical features:

- DNE: Dirichlet Normal Energy – a general measure of the tooth surface curvature.
- positive_DNE: Positive curvature part (like peaks and valleys).
- surface_area: Total surface area of the tooth.
- positive_surface_area: The area that has positive curvature.

Each sample also has a Diet label (the kind of food it eats) and Genus (the monkey type). The diet types are:

- *Frugivore* (fruit),
- *Folivore* (leaves),
- *Omnivore* (everything),
- *Insectivore* (insects),
- *Frugivore-Folivore* (fruit and leaves),
  and some have NA when the diet is unknown (usually extinct monkeys).

Is Frugivore-Folivore a separate group?

Before starting the classification, we wanted to see if the group "Frugivore-Folivore" was really different from "Folivore". If not, we could merge them to simplify the task.

We did a Hotelling's T^2 permutation test (with 9999 permutations) and the result was:

- *p-value* > *0.05*, so we cannot say they are different.
- So we decided to merge Frugivore-Folivore into Folivore.

Choosing classification method

At first we thought about using Linear Discriminant Analysis (LDA). But to check if it was a good idea, we did a Box's M test. The result showed:

- *p-value < 0.05* which means the covariance between groups is different.
- LDA assumes equal covariance, so it's not a good fit here.

Because of that, we chose these two models:

- Quadratic Discriminant Analysis (QDA): works better when covariance is different.
- Random Forest: more flexible, can capture complex patterns.

We used both models to predict the diet of *Teilhardina*, the extinct genus with unknown diet.

Data processing (our approach)

- We removed all rows with NA in diet.
- We only used the four numerical features.
- We scaled the features using *StandardScaler* from sklearn, calculated as: (mean - datapoint)/standard deviation
- We trained the models using the known diets.
- We tested the models on the 8 *Teilhardina* samples.
- 90/10 split for training and testing

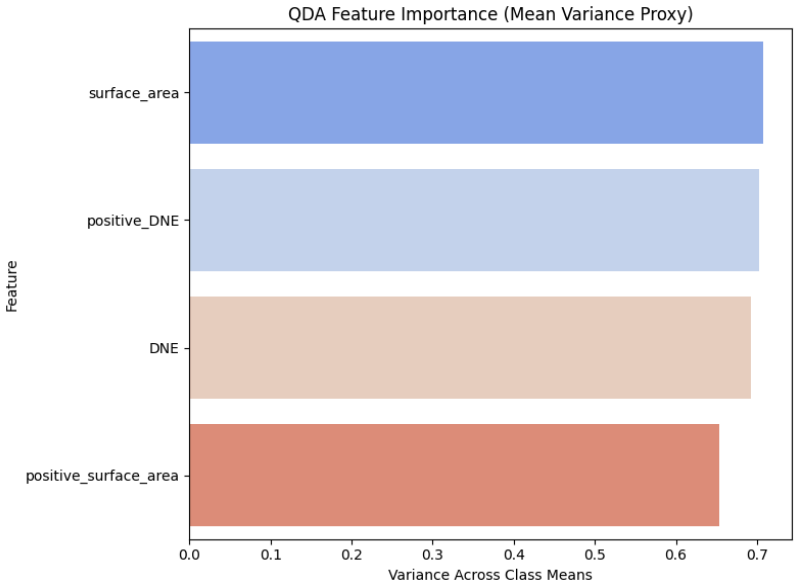
Results from our models

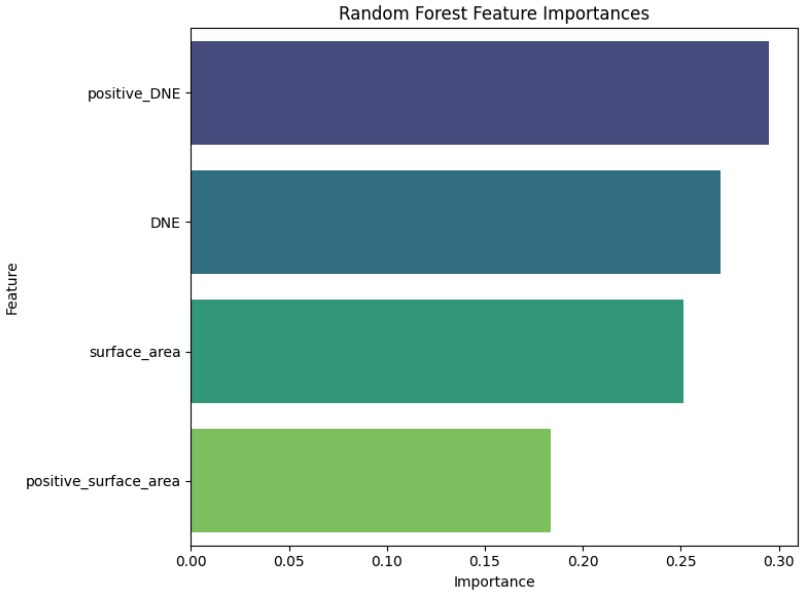
Quadratic Discriminant Analysis (QDA)	
Prediction for Teilhardina	Count
Omnivore	8
Random Forest	
Prediction for Teilhardina	Count
Omnivore	7
Folivore	1

Both models strongly suggest that Teilhardina was Omnivore. Only one tooth got classified as Folivore by Random Forest.

Discussion

- The two models agree that Teilhardina is most similar to modern Omnivores.
- The one disagreement in Random Forest could be due to normal variation or model behavior.
- The models are simple, but the dataset is also quite small. Cross-validation or more data would make it stronger.
- It should be noted that scaling is very important for this dataset since all the features do not have the same unit. Classification without scaling yields very different results.
- We determined the feature importances for both classification models which can be seen below.





- We decided against using the Genus column as a classification feature, arguing that it had the potential to affect the prediction results significantly based on how the monkeys are grouped as species.

Conclusion

We analyzed the shape of primate molars and tried to predict the diet of the extinct genus Teilhardina. Both classification models show strong agreement that Teilhardina was probably an Omnivore, which means it ate a mix of food like fruit, leaves, or insects. This gives us some insight into how this monkey may have lived and interacted with its surrounding environment.